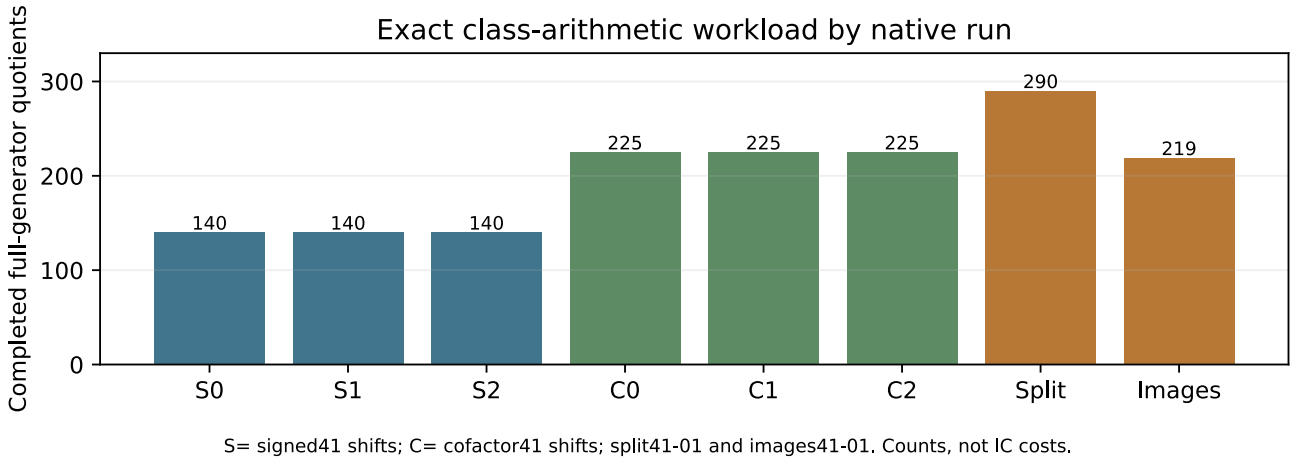
Koblitz transfers — R41

Both crossed source operators are nonzero on an independent collector

Exact native observations, 2026-10-30. Eight native runs and two Lean workflows accepted. No IC relation rank, speedup or scalar-recovery claim.

All ten complete divisor spaces of collector orbit2 are used. The actual diamond order is independently rebound. Signed classes at shifts0,1,2 are nonprincipal, with two independent expressions agreeing at every shift. Direct3721E agrees with61·(61E′); each image is nonzero and killed11.

The three actual cofactored classes satisfy T²F+F=4TF. Consequently the constructed P7=8F-TF and P8=TF-7F are both nonprincipal, killed11, recoverF, and satisfy checked actual diamond actions7P7 and8P8. These are two disjoint source characters under the retained proper/endomorphism bridge.



S= signed41 shifts; C= cofactor41 shifts; split41-01 and images41-01. Counts, not IC costs.

The full crossed integer operators reduce on this rational input to Pi7(F)=6F+2TF and Pi8(F)=3F+9TF. Actual class operations agree with independent 9P7 and9P8 constructions. Both images are nonprincipal and recovery 5·(Pi7(F)+Pi8(F))=F passes. Every one of the1604 subspace quotients uses all original generators, with independent product arithmetic.

Exhaustive comparison with all11 multiples of the saved old sourceR gives P7=9R and no P8 match. Thus the independent collector reaches a component outside the old scalar line. This is a tiny exact source-group control, not unknown-scalar recovery or an IC relation-rank measurement.

Nine Lean theorems prove conditional recovery/equivariance/disjointness implications, with zero failing and zero trusted declarations. Arrays, proper/RR/CM bridges and the chosen Koblitz model isogeny are not formalized.

The explicit chosen-target isogeny/convention binding, full saturation and polarization remain open. Larger615 source jets/spaces, degree460/367 supports, theta, node187 and actual larger-n transfers remain unfinished. Next test twist-specific residual separation at actual n11/83/131 primes, distinguishing a generator diamond from diamond2.

Native4GiBvirtual/1200s; Lean4.34.1/mathlibd13f23b,-j1-M4096/1200s. Sources: KM math/0409209v3; R15/R18 CM type; R34–R40 source/local/class certificates. See REPORT.md, NEXT.md, protocols and complete raw evidence.